Git Version Control Commands - Explanations

A) Setup & Configuration

- git --version

→ Shows the installed Git version.

- git config --global user.name "Your Name"

→ Sets your username globally.

- git config --global user.email you@example.com

→ Sets your email globally.

B) Create or Clone Repositories

- git init

→ Creates a new Git repository.

- git clone <repository-url>

→ Clones a remote repository.

C) Check Status

- git status

→ Shows file changes and staging info.

D) Add Files (Staging Area)

- git add file.txt

→ Stages a specific file.

- git add .

→ Stages all changed files.

E) Commit Changes

- git commit -m "Message"

→ Creates a new commit.

- git commit --amend

→ Modifies the previous commit.

F) View History

- git log

→ Shows full commit history.

- git log --oneline

→ Condensed commit history.

- git log --oneline --graph --all

→ Visual branch history.

G) Branching

- git branch

→ Lists branches.

- git branch <name>

→ Creates a new branch.

- git checkout <name>

→ Switches branch.

- git checkout -b <name>

→ Creates & switches to a branch.

H) Merging

- git merge <branch>

→ Merges a branch into current.

I) Remote Repository Commands

- git remote add origin <url>

→ Adds remote repo.

- git remote -v

→ Shows remotes.

- git push origin main

→ Pushes changes.

- git push -u origin main

→ Pushes & sets upstream.

- git pull origin main

→ Fetches & merges.

- git fetch

→ Fetches without merging.

J) Undo / Reset / Restore

- git restore file.txt

→ Restores file to last commit.

- git restore --staged file.txt

→ Unstages a file.

- git reset --soft HEAD~1

→ Undo commit, keep staged.

- git reset --mixed HEAD~1

→ Undo commit, keep changes.

- git reset --hard HEAD~1

→ Undo commit and delete changes.

- git revert <commit-id>

→ Creates a reverse commit.

K) Stash (Temporary Save)

- git stash

→ Saves work temporarily.

- git stash list

→ Shows all stashes.

- git stash apply

→ Applies stash.

- git stash pop

→ Applies & removes stash.

L) Tagging

- git tag v1.0

→ Creates a tag.

- git push origin v1.0

→ Pushes tag to remote.